

Workshop 1

Project: MusicLock

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1. Requirements Documentation

1.1. Functional requirements:

- **Requirements Account management and authentication**

- **FR-01:** The system shall allow new users to register an account by providing an email address, username, and password.
- **FR-02:** The system shall allow users to log in by validating their credentials and issuing a JSON Web Token (JWT) upon successful authentication.
- **FR-03:** The system shall automatically renew the user's session using a JWT refresh token, without requiring manual re-authentication.
- **FR-04:** The system shall allow users to log out, which must successfully invalidate or revoke the active JWT to terminate the session.

- **Requirements Audio Management and Encryption**

- **FR-01:** The client-side application must encrypt audio files locally before initiating the upload process to the server.
- **FR-02:** The client-side application must decrypt audio files locally during playback using the user's derived symmetric key.
- **FR-03:** The client-side application must upload only the encrypted file to the server, never the original MP3.
- **FR-04:** The system must store the encrypted file without having access to the keys required to decrypt it.
- **FR-05:** The client-side application must decrypt audio files in memory to allow the user to download the original file.

- **Requirements Interface**

- **FR-01:** The system must provide a dashboard as the main panel from which the user can access all the platform's options.
- **FR-02:** The dashboard must display a section with the files marked as favorites by the user.
- **FR-03:** The dashboard must display a section with the user's playlists.

- **FR-04:** The dashboard must display a "Shared with me" section with the files and playlists shared with the user.
 - **FR-05:** The dashboard must display a "Recently played" section with the last songs played by the user.
 - **FR-06:** The dashboard must display the storage used and the storage available according to the user's plan.
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- **Requirements Access Management and Playlist**
 - **FR-01:** The system shall allow a user to share access to a specific audio file or playlist with another designated user.
 - **FR-02:** The system shall allow a user to revoke access to a file or playlist that was previously granted to another user.
 - **FR-03:** The system must allow the user to assign a permission role (Query) to each user with whom a file is shared.
 - **FR-04:** The system must allow the user to modify the permission role of a user who already has shared access to a file.
 - **FR-05:** The system must allow the user to list all users who have access to a specific private file or playlist.
 - **FR-06:** The system must allow the user to list all private files or playlists that have been shared with the current user.
 - **FR-07:** The system must validate the current user's permissions (Query/Consulta) before allowing any action on a shared file.
 - **FR-08:** The system must allow the user to create a playlist.
 - **FR-09:** The system must allow the user to edit a playlist created by them.
 - **FR-10:** The system must allow the user to delete a playlist created by them.
 - **FR-11:** The system must allow the user to add a song to a playlist.
 - **FR-12:** The system must allow the user to remove a song from a playlist.
 - **FR-13:** The system must allow the user to list all songs marked as favorites.
 - **FR-14:** The system must provide an integrated web player to play playlist audio files decrypted directly within the browser.

- **Requirements Album System**

- **FR-01:** The system must allow the user to create an album, which will be associated with that user.
- **FR-02:** The system must allow the user to delete an album created by them.
- **FR-03:** The system must allow the user to edit an album created by them.
- **FR-04:** The system must allow the user to associate a song owned by them with an album created by them.
- **FR-05:** The system must allow the user to disassociate a song from an album created by them.
- **FR-06:** The system must allow the user to list all albums by an artist.
- **FR-07:** The system shall allow the user to list all songs of an album.

- **Requirements Social Interaction and Community**

- **FR-01:** The system must allow the user to follow another user.
- **FR-02:** The system must allow the user to unfollow a user they were previously following.
- **FR-03:** The system must allow the user to list their followers.
- **FR-04:** The system must allow the user to list the users they follow.
- **FR-05:** The system must allow the user to view the public playlists of followed users.
- **FR-06:** The system must allow the user to add a song from a public playlist of another user to their own playlist.
- **FR-07:** The system must allow the user to post a comment on a song.
- **FR-08:** The system must allow the user to post a comment on a public playlist.
- **FR-09:** The system must allow the user to edit their own comments.
- **FR-10:** The system must allow the user to delete their own comments.
- **FR-11:** The system must display the timestamp for each comment to the user.

- **Requirements Copyright protection for public Files**

- **FR-01:** The system must verify, through an external service such as ACRCLOUD, whether a public file has copyright before allowing it to be published.
- **FR-02:** If a public file has copyright, the system must prevent its publication and notify the user responsible for uploading the file.

- **Requirements Statistics and Analysis**

- **FR-01:** The system must record each playback, including the user, song, date, and time, in a playback history.
- **FR-02:** The system must calculate the top most-played songs from the last 7 days.
- **FR-03:** The system must calculate the top most-played songs from the last 30 days.
- **FR-04:** The system must calculate the top most-listened artists for each user.
- **FR-05:** The system must display the total number of plays for each song.
- **FR-06:** The system must display a user's playback history, including their most recently played songs.
- **FR-07:** The system must generate a general trending ranking of the most-played songs on the platform.

- **Requirements Premium Functionality**

- **FR-01:** The system must define the subscription plans (Free and Premium).
- **FR-02:** The system must assign a storage limit according to the user's plan (e.g., Free: 500 MB, Premium: 5 GB).
- **FR-03:** The system must restrict features according to the user's plan (e.g., maximum number of playlists or uploaded songs on the Free plan; multiple libraries on the Premium plan).
- **FR-04:** The system must allow the user to upgrade or downgrade their subscription plan.
- **FR-05:** The system must validate the available storage space before allowing the upload of new files.
- **FR-06:** The system must allow the administrator to manage the Premium plans and review the status of users' subscriptions.

- **Requirements Scheduled decryption**

- **FR-01:** The system must allow the user to enable the decryption of their files to simulate the release of a song.
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1.2. Non-Functional requirements:

- **Security**

- **NFR-01:** The backend must never store passwords in plain text.
- **NFR-02:** The client-side application must generate a master key for each user using the native web Crypto API, protect it with a key derived from the user's password, and store only the protected version, so the user can recover it from any device.
- **NFR-03:** The encryption and decryption of MP3 audio files must be performed exclusively in the user's browser (Client-Side Encryption) using the native Web Crypto API (with AES-GCM) before being transmitted or stored in the cloud.
- **NFR-04:** The local encryption key must be derived from the user's password using the Argon2id algorithm (or PBKDF2 depending on the frontend/backend implementation) to prevent brute-force attacks and ensure the confidentiality of the master key.
- **NFR-05:** Requests to the REST API must be authenticated using JWT tokens (JSON Web Tokens). The platform must support token renewal via refresh tokens without exposing sensitive data or requiring constant user re-authentication.

- **Architecture**

- **NFR-01:** Backend authentication must be implemented using Django's built-in authentication system and a standard JWT implementation library.
- **NFR-02:** The client-side cryptographic operations and UI must be implemented using React and Typescript.

- **Privacy**

- **NFR-01:** (Zero-Knowledge Privacy / Confidentiality): Neither the server, the database (PostgreSQL), system administrators, nor object storage must have access to the keys to decrypt the content of the files.

- **Response Time**

- **NFR-01:** The system must respond to navigation and data read requests (such as querying playlists, profiles, or comments) within a maximum of 2 seconds, and process write operations (such as posting comments or following a user) in no more than 1 second under normal load conditions.
- **NFR-02:** Copyright verification via the ACRCLOUD API must run asynchronously or be optimized to respond within a few seconds before allowing the publication of a public file.
- **NFR-03:** The local in-browser decryption process (in-memory) for playback or download must not introduce noticeable UI latency or delay, ensuring smooth web player playback.

- **Availability**

- **NFR-01:** The system must guarantee 99.9% uptime (24/7/365), allowing for a maximum unplanned downtime of no more than 8.76 hours per year.
- **NFR-02:** The entire application (Frontend in React+TypeScript, Backend in Django, and PostgreSQL) must be containerized and isolated using Docker to ensure environment consistency and facilitate scaling on cloud infrastructure.

- **Scalability**

- **NFR-01:** The backend architecture must scale to support massive loads and encrypted blob storage, enforcing tiered storage quotas (e.g., 500 MB for Free plans and 5 GB for Premium plans).

- **Portability**

- **NFR-01:** The system must allow users to authenticate and log in from any web-enabled device, seamlessly recovering or deriving their keys without requiring complex technical configurations.

- **Legality**
 - **NFR-01:** The system must prevent the public publication of content that violates copyright through strict integration with the ACRCLOUD identification API.

2. User Stories

1. Epic 1: Account management and authentication

- **UH-01:** As a prospective user, I want to register a new account using my email, username, and password so that I can access the platform and secure my personal content.
 - **Acceptance Criteria:**

Given that the user is on the registration page, when they submit a valid email address, username, and password, then the system creates the account, securely hashes the password using Django's hashing system, and confirms successful registration.
- **UH-02:** As a registered user, I want to log in using my account credentials so that I can authenticate securely and access my protected resources.
 - **Acceptance Criteria:**

Given that the user is on the login page, when they enter valid credentials, then the system validates the credentials, derives the client-side key, issues a valid JSON Web Token (JWT), and grants access to the platform.
- **UH-03:** As an authenticated user, I want my session to automatically renew in the background using a refresh token so that I can maintain an uninterrupted experience without manually re-authenticating.
 - **Acceptance Criteria:**

Given that the user has an active session and an expiring JWT access token, when a background request is triggered with a valid refresh token, then the system issues a new JWT access token without prompting the user for credentials.

- **UH-04:** As a logged-in user, I want to log out of the system so that my active session is terminated and my token cannot be reused.

- **Acceptance Criteria:**

Given that the user is logged into the platform, when they click the "Log Out" button, then the system invalidates/revokes the active JWT, clears local session storage, and redirects the user to the login screen.

2. Epic 2: Audio Management and Encryption

- UH-01:** As a registered user, I want to have my MP3 files encrypted in my browser before they are uploaded so that no one, including the server, can access my music.

- Acceptance Criteria:**

Given that the user is logged into the platform and selects an MP3 file, when the user confirms the upload, then the client encrypts the file locally with AES-GCM and sends only the encrypted file to the server.

- UH-02:** As a registered user, I want my encrypted files to be stored without the platform having the keys so that neither administrators nor the database can decrypt my content.

- Acceptance Criteria:**

Given that the server has received an encrypted file, when the upload finishes, then the system stores the file linked to the user's account and never stores or receives the decryption key.

- UH-03:** As a registered user, I want to download the decrypted original version of my song so that I can use my music outside the platform.

- **Acceptance Criteria:**

Given that the user is logged into the platform and owns an encrypted song, when the user selects the download option, then the client decrypts the file in memory and delivers the original MP3 to the user's device.

- UH-04:** As a registered user, I want to access my music from any device after logging in so that I don't have to manage keys manually.

- Acceptance Criteria:**

Given that the user uploaded encrypted files from one device, when the user logs in with valid credentials from another device, then the client recovers the user's keys and can decrypt and play those files.

- **UH-05: As a free user, I want the system to validate whether I have exceeded my 500 MB limit before letting me upload more music so that I know my available space and don't waste time uploading a file that will be rejected.**

- **Acceptance Criteria:**

Given that the user is on the Free plan, **when** the size of the selected file plus the space already used exceeds 500 MB, **then** the system blocks the upload before it starts, shows the space used and available, and suggests upgrading to Premium.

UH-06: As a registered user, I want to upload an MP3 file so that I can save it on the platform and listen to it later.

- **Acceptance Criteria:**

Given that the user is logged into the platform, **when** the user selects a valid MP3 file and confirms the upload, **then** the system stores the file, associates it with the user's account and shows it in their library. If the file is not a valid MP3, the system rejects it and shows an error message.

UH-07: As a registered user, I want to play, pause, adjust the volume and repeat a song so that I can control my listening experience directly in the browser.

- **Acceptance Criteria:**

Given that the user is logged into the platform and a song is loaded in the web player, **when** the user presses play or pause, changes the volume or activates repeat, **then** the player responds immediately: playback starts or stops, the volume changes, and the song restarts when it ends if repeat is on.

3. Epic 3: Interface

- **UH-01: As a registered user, I want to access a dashboard with all the platform's options so that I can navigate to any feature from a single place.**

- **Acceptance Criteria:**

Given that the user is logged into the platform, **when** the user enters the main

page, **then** the system displays the dashboard with the sections Favorites, Playlists, Shared with me, Recently played and Storage.

- **UH-02: As a registered user, I want to** see my favorites, playlists and shared content in the dashboard **so that** I can reach my music quickly without searching through menus.
 - **Acceptance Criteria:**

Given that the user is logged into the platform, **when** the dashboard loads, **then** the Favorites, Playlists and Shared with me sections show the items that correspond to that user (as defined in the Access Management and Playlists stories).
- **UH-03: As a registered user, I want to** see the songs I played recently **so that** I can resume listening to them quickly.
 - **Acceptance Criteria:**

Given that the user is logged into the platform and has played songs before, **when** the user opens the "Recently played" section, **then** the system displays the last songs played, ordered from most to least recent.
- **UH-04: As a registered user, I want to** see how much storage I have used and how much is free **so that** I can manage my files before reaching my plan's limit.
 - **Acceptance Criteria:**

Given that the user is logged into the platform, **when** the dashboard loads, **then** the system displays the used and free storage, calculated against the limit of the current plan of the user.

4. Epic 4: Access Management and Playlists

- **UH-01: Create, Edit a playlist:** As a registered user, I want to create and edit a playlist so that I can start grouping my favorite audio tracks.
 - **Acceptance Criteria:**
- Given that the user is logged into the platform, when the user inputs a valid playlist name and triggers the creation process, then the system generates a new playlist record in the database and links it exclusively to the user's account.
- **UH-02: Add a song to a playlist:** As a playlist owner, I want to add a song to my playlist so that I can expand my curated audio collection.

- **Acceptance Criteria:**
Given that the user is logged into the platform and is viewing a playlist they own, when the user selects a song and chooses to add it to that playlist, then the system links the song to the playlist record and updates the tracklist view.
- **UH-03: List favorite songs:** As a registered user, **I want to** list all the songs I have marked as favorites **so that** I can quickly access my preferred music from a single view.
 - **Acceptance Criteria:**
Given that the user is logged into the platform, **when** the user navigates to the "Favorites" or "Liked Songs" section, **then** the system filters and displays only the tracks that have been flagged as a favorite by that specific user.
- **UH-04: Delete a playlist:** As a playlist owner, **I want to** delete a playlist created by me **so that** I can remove collections I no longer listen to.
 - **Acceptance Criteria:**
Given that the user is logged into the platform and owns an existing playlist, **when** the user selects the option to delete that playlist and confirms the action, **then** the system permanently removes the playlist from the user's account.
- **UH-05: Remove a song from a playlist:** As a playlist owner, **I want to** remove a song from my playlist **so that** I can keep my collection clean and updated.
 - **Acceptance Criteria:**
Given that the user is logged into the platform and is viewing a playlist they own, **when** the user selects a song inside the playlist and clicks on the remove option, **then** the system deletes the association between that song and the playlist.
- **UH-06: As a file owner, I want to** assign a permission role (Query) to a user with whom I share a file **so that** I can control what actions they can perform on my shared content.
 - **Acceptance Criteria:**
Given that the user is logged into the platform and is the owner of a file, **when** the user shares the file and selects “private”, it shares the file with another user and selects the "Query" permission role, **then** the system updates the sharing permissions and grants the specified access to that target user.

Given that the user is logged into the platform and is the owner of a file, **when** the user shares a file and selects “public”, **then** it gives the "Query" permission role to every user in the platform.

- **UH-07: As a file owner, I want to** modify the permission role of a user who already has access to my shared file **so that** I can dynamically change their access rights when needed.

- **Acceptance Criteria:**

Given that the user is logged into the platform and owns a shared file, **when** the user selects an existing shared user and changes their permission level, **then** the system updates the permission matrix in the database and applies the new access rights instantly.

- **UH-08: As a file or playlist owner, I want to** list all the users who currently have access to that resource when it is private **so that** I can audit and keep track of who can view my content.

- **Acceptance Criteria:**

Given that the user is logged into the platform and views the details of a file or playlist owned by them, **when** the user opens the "Shared with" or access management section, **then** the system retrieves and displays a comprehensive list of all users with active access along with their respective role, in case that the file was shared as “private”.

Given that the user is logged into the platform and views the details of a file or playlist owned by them, **when** the user opens the "Shared with" or access management section, **then** if the file was shared as “public” the system will show a message that says “shared with everyone”.

- **UH-09: As a registered user, I want to** list all the files or playlists that other users have shared with me **so that** I can easily access content that doesn't belong to me but I have permission to view.

- **Acceptance Criteria:**

Given that the user is logged into the platform and navigates to the "Shared with me" section, **when** the page loads, **then** the system fetches and lists all

files and playlists where the current user's ID is explicitly associated with a shared permission role.

5. Epic 5: Album System

- **UH-01:** As a registered user, I want to create an album so that I can organize my songs under a specific collection associated with my account.
 - **Acceptance Criteria:**

Given that the user is logged into the platform and is on the album creation page, when the user enters an unique album title and submits the creation form, then the system creates the new album on the database and associates it directly with the user's account.
- **UH-02:** As a registered user, I want to delete an album created by me so that I can remove collections I no longer need from my account.
 - **Acceptance Criteria:**

Given that the user is logged into the platform and owns an existing album, when the user selects the option to delete that album and confirms the deletion, then the system permanently removes the album from the user's account.
- **UH-03:** As a registered user, I want to edit the cover, title and description of an album created by me so that I can update its details and keep my collection accI want to delete accurate.
 - **Acceptance Criteria:**

Given that the user is logged into the platform and owns an existing album, when the user updates the album details and submits the changes, then the system saves the updated information for that album.
- **UH-04:** As a registered user, I want to associate a song owned by me with an album created by me so that I can organize my music into specific collections.
 - **Acceptance Criteria:**

Given that the user is logged into the platform and owns an existing album, when the user selects a song owned by them and assigns it to that album, then the system links the song to the selected album.
- **UH-05:** As a registered user, I want to disassociate a song from an album created by me so that I can remove it from that collection without deleting the song.

- **Acceptance Criteria:**

Given that the user is logged into the platform and has a song associated with an album they own, when the user selects the option to remove the song from the album, then the system unlinks the song from that album.

- **UH-06:** As a user, I want to list all albums by a specific artist so that I can explore their available music collection easier.

- **Acceptance Criteria:**

Given that the user is browsing an artist's profile or catalogue, when the user selects the option to view albums by that artist, then the system displays a list of all albums associated with the selected artist.

- **UH-07:** As a user, I want to list all songs of an album so that I can view and access the full track listing of that collection.

- **Acceptance Criteria:**

Given that the user is viewing a specific album, when the user opens the album details page, then the system displays all songs associated with that album.

6. Epic 6: Social Interaction and Community

- **UH-01:** As a registered user, I want to follow another user so that I can stay updated with their activity and public content.

- **Acceptance Criteria:**

Given that the user is logged into the platform and is viewing another user's profile, when the user clicks the option to follow that user, then the system establishes the “follow” relationship and updates the user's list of followed accounts.

- **UH-02:** As a registered user, I want to unfollow a user I am currently following so that I no longer see their updates or public content.

- **Acceptance Criteria:**

Given that the user is logged into the platform and is currently following another user, When the user selects the option to unfollow that user, Then the system removes the follow relationship and updates the user's list of followed accounts.

- **UH-03:** As a registered user, I want to list my followers so that I can see which users are following my account and public updates.

- **Acceptance Criteria:**
Given that the user is logged into the platform, when the user accesses their profile follower section, then the system displays the complete list of users who currently follow their account.
- **UH-04:** As a registered user, I want to list the users I follow so that I can view and access the accounts I have previously chosen to connect with.
 - **Acceptance Criteria:**
Given that the user is logged into the platform, When the user accesses the section of accounts they follow, Then the system displays the complete list of users they currently follow.
- **UH-05:** As a registered user, I want to view the public playlists of users I follow so that I can discover new music and listen to their shared collections.
 - **Acceptance Criteria:**
Given that the user is logged into the platform and follows another user, When the user accesses the followed user's profile or playlist section, Then the system displays all playlists marked as public by that user.
- **UH-06:** As a registered user, I want to add a song from another user's public playlist to my own playlist so that I can easily save tracks I discover while browsing other users' collections.
 - **Acceptance Criteria:**
Given that the user is logged into the platform and viewing a public playlist belonging to another user, when the user selects a song from that playlist and chooses the option to add it to one of their own playlists, then the system adds the selected song to the user's chosen playlist.
- **UH-07:** As a registered user, I want to post a comment on a song so that I can share my thoughts and engage with the community about that track.
 - **Acceptance Criteria:**
Given that the user is logged into the platform and viewing a specific song, when the user enters a comment and submits it, then the system saves the comment and displays it under the song's comment section.
- **UH-08:** As a registered user, I want to post a comment on a public playlist so that I can share my opinion and engage with the owner and other listeners.
 - **Acceptance Criteria:**

Given that the user is logged into the platform and viewing a public playlist, when the user enters a comment and submits it, then the system saves the comment and displays it under the public playlist's comment section.

- **UH-09:** As a registered user, I want to edit my own comments so that I can correct mistakes or update my thoughts on a song or public playlist.
 - **Acceptance Criteria:**

Given that the user is logged into the platform and is the author of an existing comment, when the user edits the comment text and submits the changes, then the system updates the comment and displays the new text in the comment section.
- **UH-10:** As a registered user, I want to delete my own comments so that I can remove my thoughts or feedback from a song or public playlist.
 - **Acceptance Criteria:**

Given that the user is logged into the platform and is the author of an existing comment, when the user selects the option to delete that comment and confirms, then the system permanently removes the comment from the comment section.
- **UH-11:** As a user, I want to see the timestamp for each comment so that I know when a comment was originally posted or updated.
 - **Acceptance Criteria:**

Given that the user is viewing the comment section of a song or public playlist, when the comments are displayed, then the system shows the associated creation or update timestamp alongside each comment.

7. Epic 7: Copyright Protection for Public Files

- **UH-1:** As a user, I want the system to verify whether a public file has copyright through an external service such as ACRCLOUD, so that the file can be checked before being published.
 - **Acceptance Criteria:**

Given that the user wants to publish a public file, when the system verifies the file through an external service such as ACRCLOUD, then the system determines whether the file has copyright.

- **HU-02:** As a user, I want the system to prevent a public file with copyright from being published and notify the responsible user, so that the copyrighted file cannot be published.

- **Acceptance Criteria:**

Given that a public file has copyright, when the user attempts to publish it, then the system prevents its publication and notifies the user responsible for uploading the file.

8. Epic 8: Statistics and Analysis

- **UH-01:** As a user, I want each playback to be recorded with the user, song, date, and time, so that a playback history can be maintained.

- **Acceptance Criteria:**

Given that a user plays a song, when the playback is registered, then the system records the user, song, dPlayback Statistics and Trends ate, and time in the playback history.

- **UH-02:** As a user, I want to see the top most-played songs from the last 7 days, so that I can see which songs have been most played during that period.

- **Acceptance Criteria:**

Given that playback records from the last 7 days exist, when the system calculates the ranking, then it displays the top most-played songs from that period.

- **UH-03:** As a user, I want to see the top most-played songs from the last 30 days, so that I can see which songs have been most played during that period.

- **Acceptance Criteria:**

Given that playback records from the last 30 days exist, when the system calculates the ranking, then it displays the top most-played songs from that period.

- **UH-04:** As a user, I want to see my most-listened artists, so that I can know which artists I listen to the most.

- **Acceptance Criteria:**

Given that the user has playback records, when the system calculates the user's most-listened artists, then it displays the artists according to their number of plays.

- **UH-05:** As a user, I want to see the total number of plays for each song, so that I can know how many times each song has been played.

- **Acceptance Criteria:**

Given that a song has playback records, when the user views the song information, then the system displays its total number of plays.

- **UH-06:** As a user, I want to view my playback history, so that I can see the songs I have recently listened to.

- **Acceptance Criteria:**

Given that the user has listened to songs, when the user accesses their playback history, then the system displays their most recently played songs.

- **UH-07:** As a user, I want to see a general ranking of the most-played songs on the platform, so that I can see the current trends of the platform.

- **Acceptance Criteria:**

Given that playback records exist on the platform, when the system generates the general trending ranking, then it displays the most-played songs on the platform.

9. Epic 9: Premium Functionality

- **UH-01:** As a system administrator, I want to define and manage the subscription plans (Free and Premium) with their storage limits so that I can offer different service levels to users.

- **Acceptance Criteria:**

Given that the administrator is logged into the platform, when the administrator creates or edits a plan, then the system saves its name, storage limit (e.g., 500 MB / 5 GB) and feature restrictions, and applies them to the users of that plan.

-**UH-02:** As a system administrator, I want to restrict features according to the user's plan **so that** extended benefits (higher limits, multiple libraries) remain exclusive to Premium users.

- Acceptance Criteria:**

Given that a Free user has reached the maximum number of playlists or uploaded songs allowed, when the user tries to create another one, then the system blocks the action and invites the user to upgrade to Premium.

-**UH-03:** As a registered user, I want to upgrade or downgrade my subscription plan so that I can adapt the service to my needs.

- Acceptance Criteria:**

Given that the user is logged into the platform, when the user selects a different plan and confirms, then the system updates the plan and applies the new limits immediately. If a downgrade would leave the user over the new

storage limit, the system warns the user and blocks the change until enough space is freed.

10. Epic 10: Scheduled decryption

UH-01: As a user, I want to enable the decryption of my files, so that I can simulate the release of a song.

Acceptance Criteria:

Given that the user has a file, when the user enables the decryption of the file, then the system allows the file to be decrypted to simulate the release of a song.

3. User Stories Map



Epicas	Epic 1: Account management and authentication			
Actividades	Register Account		Authenticate JWT	Derive Local Keys
HU	UH-01: As a prospective user, I want to register a new account using my email, username, and password so that I can access the platform and secure my personal content		UH-02: As a registered user, I want to log in using my account credentials so that I can authenticate securely and access my protected resources.	
			UH-04: As a logged-in user, I want to log out of the system so that my active session is terminated and my token cannot be reused.	
Release 1				
HU				UH-03: As an authenticated user, I want my session to automatically renew in the background using a refresh token so that I can maintain an uninterrupted experience without manually re-authenticating.
Release 2				
HU				
Release 3				
HU				
HU				
Release 4				



Epic 2. Audio Management and Encryption		
Upload Encrypted Audio	Validate Storage	Stream Playback
UH-6As the owner, I want to upload an MP3 file to save it on the platform" (Plain unencrypted stream to test that Django receives files).		UH-01: As a registered user, I want to have my MP3 files encrypted in my browser before they are uploaded so that no one, including the server, can access my music
		UH-03: As a registered user, I want to download the decrypted original version of my song so that I can use my music outside the platform.
		UH-07As a user, I want to play, pause, adjust the volume, and repeat a song to control my audio experience directly in the browser.
Release 1		
		UH-02: As a registered user, I want my encrypted files to be stored without the platform having the keys so that neither administrators nor the database can decrypt my conten
		UH-04: As a registered user, I want to access my music from any device after logging in so that I don't have to manage keys manually.
Release 2		
	UH-05: As a free user, I want the system to validate whether I have exceeded my 500 MB limit before letting me upload more music so that I know my available space and don't waste time uploading a file that will be rejected.	
Release 3		



Release 4		



Epic 3. Interface				
Access the dashboard		Browse content		Monitor storage
UH-01: As a registered user, I want to access a dashboard with all the platform's options so that I can navigate to any feature from a single place.				UH-04: As a registered user, I want to see how much storage I have used and how much is free so that I can manage my files before reaching my plan's limit.
Release 1				
Release 2				
		UH-02: As a registered user, I want to see my favorites, playlists and shared content in the dashboard so that I can reach my music quickly without searching through menus.		
Release 3				



		UH-03: As a registered user, I want to see the songs I played recently so that I can resume listening to them quickly.		
Release 4				



Epic 4: Access Management and Playlists		
Organizar en playlists	Marcar Favoritos	Compartir Permisos
UH-01: As a registered user, I want to create and edit a playlist so that I can start grouping my favorite audio tracks.	UH-3: List favorite songs: As a registered user, I want to list all the songs I have marked as favorites so that I can quickly access my preferred music from a single view.	
UH-02: As a playlist owner, I want to add a song to my playlist so that I can expand my curated audio collection.		
Release 1		
UH-04: Delete a playlist: As a playlist owner, I want to delete a playlist created by me so that I can remove collections I no longer listen to.		UH-06: -As a file owner, I want to assign a permission role (Query) to a user with whom I share a file so that I can control what actions they can perform on my shared content.
UH-5: Remove a song from a playlist: As a playlist owner, I want to remove a song from my playlist so that I can keep my collection clean and updated.		UH-7: As a file owner, I want to modify the permission role of a user who already has access to my shared file so that I can dynamically change their access rights when needed.
Release 2		
		UH-8: As a file or playlist owner, I want to list all the users who currently have access to that resource so that I can audit and keep track of who can view my content.
Release 3		
		UH-9: As a registered user, I want to list all the files or playlists that other users have shared with me so that I can easily access content that doesn't belong to me but I have permission to view.
Release 4		



Epic 5: Album System
Album management
Release 1
UH-01: As a registered user, I want to create and manage albums so that I can organize my songs under a specific collection associated with my account.
UH-04: As a registered user, I want to associate a song owned by me with an album created by me so that I can organize my music into specific collections.
Release 2
UH-06: As a user, I want to list all albums by a specific artist so that I can explore their available music collection easier.
UH-05: As a registered user, I want to disassociate a song from an album created by me so that I can remove it from that collection without deleting the song.
Release 3
UH-07: As a user, I want to list all songs of an album so that I can view and access the full track listing of that collection.
Release 4



Epic 6: Social Interaction and Community	
Follow Users	Comments
Release 1	
Release 2	
UH-01: As a registered user, I want to follow another user so that I can stay updated with their activity and public content. UH-05: As a registered user, I want to view the public playlists of users I follow so that I can discover new music and listen to their shared collections.	UH-07: As a registered user, I want to post a comment on a song so that I can share my thoughts and engage with the community about that track. "UH-09: As a registered user, I want to edit my own comments so that I can correct mistakes or update my thoughts on a song or public playlist. UH-10: As a registered user, I want to delete my own comments so that I can remove my thoughts or feedback from a song or public playlist"
UH-03: As a registered user, I want to list my followers so that I can see which users are following my account and public updates.	UH-11: As a user, I want to see the timestamp for each comment so that I know when a comment was originally posted or updated.
Release 3	
UH-06: As a registered user, I want to add a song from another user's public playlist to my own playlist so that I can easily save tracks I discover while browsing other users' collections.	
Release 4	



Epic 7. Copyright Protection for Public Files	
Copyright	Prevent Publication
Release 1	
UH-01As a user, I want the system to verify whether a public file has copyright through an external service such as ACRCLOUD, so that the file can be checked before being published.	UH-02;As a user, I want the system to prevent a public file with copyright from being published and notify the responsible user, so that the copyrighted file cannot be published.



Epic 8. Statistics and Analysis			
Playback Recording	Playback History	Song Statistics	Rankings and Trends
Release 1		Release 1	
UH-01: As a user, I want each playback to be recorded with the user, song, date, and time, so that a playback history can be maintained	UH-06: As a user, I want to view my playback history, so that I can see the songs I have recently listened to.	US-05: As a user, I want to see the total number of plays for each song, so that I can know how many times each song has been played.	
Release 2		Release 2	
	UH-04: As a user, I want to see my most-listened artists, so that I can know which artists I listen to the most.		UH-02: As a user, I want to see the top most-played songs from the last 7 days, so that I can see which songs have been most played during that period. US-03: As a user, I want to see the top most-played songs from the last 30 days, so that I can see which songs have been most played during that period.
			US-07: As a user, I want to see a general ranking of the most-played songs on the platform, so that I can see the current trends of the platform.



Epic 9.Premium Functionality
Premium Functionality
Release 1
Release 2
Release 3
UH-01: As a system administrator, I want to define and manage the subscription plans (Free and Premium) with their storage limits so that I can offer different service levels to users.
UH-02: As a system administrator, I want to restrict features according to the user's plan so that extended benefits (higher limits, multiple libraries) remain exclusive to Premium users.
UH-03: As a registered user, I want to upgrade or downgrade my subscription plan so that I can adapt the service to my needs
Release 4



Epic 10. Scheduled decryption
Scheduled Decryption
Release 1
US-01: As a user, I want to enable the decryption of my files, so that I can simulate the release of a song.
Release 2
Release 3
Release 4

Responsible AI Usage Statement

Artificial intelligence tools were utilized in a responsible and complementary manner to support translation, refine technical phrasing, and standardize the structure of requirements across this document. All core functional concepts, business rules, and technical constraints were defined, validated, and reviewed by human contributors to guarantee factual accuracy, contextual alignment, and system integrity. No confidential data, private credentials, or proprietary assets were shared or processed during these assistance workflows.

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